d. Accrescent: Calyx that is persistent, grows along with the fruit and encloses the fruit either completely or partially. Example: *Physalis*, Palmyra.



Figure 4.19: (d)
Persistant calyx



Figure 4.19: (e)
Accrescent

3. Shapes of calyx

Have you noticed the shoe flower's calyx? It is bell shaped called **Campanulate**. The fruiting calyx is urn shaped in *Withania* and it is called **urceolate**. In *Datura* calyx is tube like and it is known as **tubular**. Two lipped calyx is present *in Ocimum*. Sometimes calyx is coloured and called **petaloid.** Example: *Saraca*, *Sterculia*. Calyx is distinctly leafy,large and often yellow or orange coloured sometimes white as in *Mussaenda*. It is modified into hair like structure or scaly called **pappus** as in *Tridax* of Compositae.



Figure 4.20: (c)
Mussaenda



Figure 4.20: (a)
Companulate



Figure 4.20: (b)
Pappus

What is the use of pappus ?

4.3.5 Corolla

Corolla is the most attractive part in majority of the flowers and is usually brightly coloured. Corolla helps to display the flower and attracts the pollinators.

1. Fusion:

a. **Apopetalous (polypetalous, choripetalous)**: Petals are distinct. Example: *Hibiscus*.

b. **Sympetalous (gamopetalous)**: Petals are fused. Example: *Datura*.

2. Shapes of corolla

I. Apopetalous Actinomorphic

1. Cruciform: Four petals arranged in the form of a cross. Example: *Brassica*, mustard, radish, cauliflower.

2. Caryophyllaceous: Five petals with long claws with limb at right angles to the claw. Example: Caryophyllaceae. *Dianthus.*

3. Rosaceous: Five to many sessile or minutely clawed petals with radiating limbs. Example: Rose, Tea.



Figure 4.21: (a)
Cruciform



Figure 4.21: (b)
Caryophyllaceous

II. Apopetalous Zygomorphic

1. Papilionaceous:

Made up of five distinct petals organized in a butterfly shape. Corolla has three types of petals. One usually large posterior petal called **vexillum**(standard)two lateral petals-wings (**alae**) and two anterior sympetalous petals called **carina**. Example: *Clitoria ternatea*, Pea, Bean.